

Test outcome	Outcome data will be compared to intended outcome for each coding path, the results will be used to guide any alterations and improvements to each code that are required to ensure that each code pathway leads to the required outcome
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Task 2 Developing the System Prototype

From the examination of the sources and assets gathered, I have considered my options.

As the user requirements are clear and, due to legal and ethical implications of using the identified assets, all of the images are to be sourced from free imaging providers or providers that allow use for non-commercial: "License: Non-commercial Use". Some of these providers require you to set up an account to download the images and some ask you to reference the source of the image.

Care was taken when selecting the images to use due to the legal and ethical implications of the assets.

Great care was taken when selecting the external links to the health and fitness and food external links. All of the recipes were from a reputable source and promoted healthy eating and most identified the nutritional value of the meal. They were easy to follow with clear images of the prepared dishes.

The health and exercise video and information links were sponsored medical and personal trainers and supply website help and guidance as the customers will be using these videos unsupervised by health and medical experts. All of the links hosted fitness videos of a very professional quality and level. From the research of possible code snippets, the only code that I am going to modify is "chat room and blog example code" by using bootstrap which is excellent for events and layouts.

Legal and ethical consideration are important during the development of this artefact. The dilemma can be displaying people who may be overweight which reinforces that overweight means unhappy, eating lots of food and no exercise. And while this may be true to some extent, the different ways people view overweight is shocking, and this is most prevalent in the way the media portrays celebrities. Even that thin is therefore healthy. So, it is important that ethical and moral issues are thought of before selecting the images and videos.

There is a close relationship between law and ethics but not everything that is legal is ethical. Frequently law ... attempts only to set the minimum acceptable standard. The aspirations of ethical practice are higher ... It can never be appropriate to defend proposed practice solely on the basis that it is legal

Assets selected and rational:

Image	Source	Rational
	https://www.freepik.com/free-icon/search_788138.htm	Users see this search image more than the other option and it is also uses less colours.
	https://icons8.com/icons/set/settings	Will use the traditional settings icon as all users know what it is.
	https://imgbin.com/png/wfLgfRPC/computer-icons-mobile-phones-png	This is my option for "no access to member's area". Users may find the other option confusing and it has too many colours.
	https://imgbin.com/png/Jm6mWH7T/a-logo-png	Decided to use this icon as it's free for non-commercial use as the other option needs an extended license. It also stands out.
	https://www.flaticon.com/free-icon/facebook-logo-button_69407	Used this social media logo as users will be able to identify it quickly.

	https://www.stickpng.com/img/icons-logos-emojis/tech-companies/whatswebsite-logo	Used this social media logo as users will be able to identify it quickly.
	https://www.stickpng.com/img/icons-logos-emojis/tech-companies/twitter-logo	Used this social media logo as users will be able to identify it quickly.
	https://www.stickpng.com/fr/img/icones-logos-emojis/societes-de-technologie/logo-youtube-play	Used this social media logo as users will be able to identify it quickly.
	https://www.flaticon.com/free-icon/instagram-logo_69366	Used this social media logo as users will be able to identify it quickly.
Contact us example code bootstrap	This is the only code that I am going to modify. I am going to develop my code for the Website to ensure that it efficient and robust.	

Possible

Links to external sites:

https://www.bbcgoodfood.com/recipes/collection/healthy-dinner	BBC good fool link	Decided to link to these external resources, they are all verified by the medical profession and have excellent reviews on social media.
https://lovefoodhatewaste.com/	Love food healthy recipes	
https://www.ketocustomplan.com/	Kenco diet plan	
https://www.foodsavvy.org.uk/	Foodsavvy healthy recipes	
https://www.delish.com/cooking/	healthy recipes	
https://ifoodreal.com/clean-eating-recipes-dinners/	healthy recipes	The recipes featured have an image of the

https://bitesofwellness.com/	healthy recipes	food and clear recipe and methods. Most also have user reviews for each of the recipe. They are reliable sources of information to link to.
https://www.everydayhealth.com/	Health and fitness websites	
https://www.health.com/fitness	Health and fitness websites	
https://www.health.harvard.edu/topics/exercise-and-fitness	Health and fitness websites	
https://www.pinterest.com/reachyourpeak/health-fitness-topics/	Health and fitness websites	

Blogs:

https://www.quora.com/What-are-your-top-5-favorite-health-and-fitness-topics	Fitness blog	Will provide further options for the customers to select their own topics
https://www.wix.com/	Wix blog	These are the top 3 blogs in the sector and have been established for some time and have a good following and lots of posts.
https://bitesofwellness.com/blog/	Bites of Wellness	
https://blog.feedspot.com/uk_fitness_blogs/	Top 10 blogs	

Prototype for the proposed digital system:

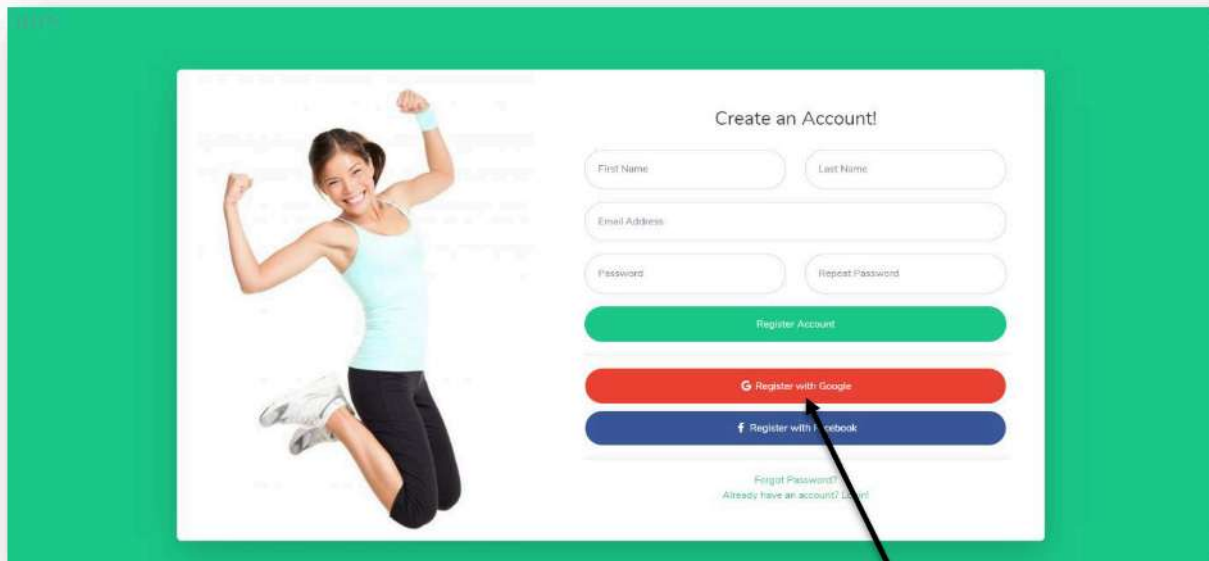
ToKa Fitness system prototype has commissioned my software development company to develop a digital system. ToKa Fitness offers personal training sessions and advice on fitness training and healthy living to its customers and would like a Website for their specific requirements.

ToKa Fitness specific requirements are that the proposed solution.

- has free access with some accessibility to services
- customers section for paid content to full services
- accessibility features for users with sight loss
- link to 'social media' features
- ability to customizable workout and eating plans
- Access to various graphs to view stats on;
 - Calories

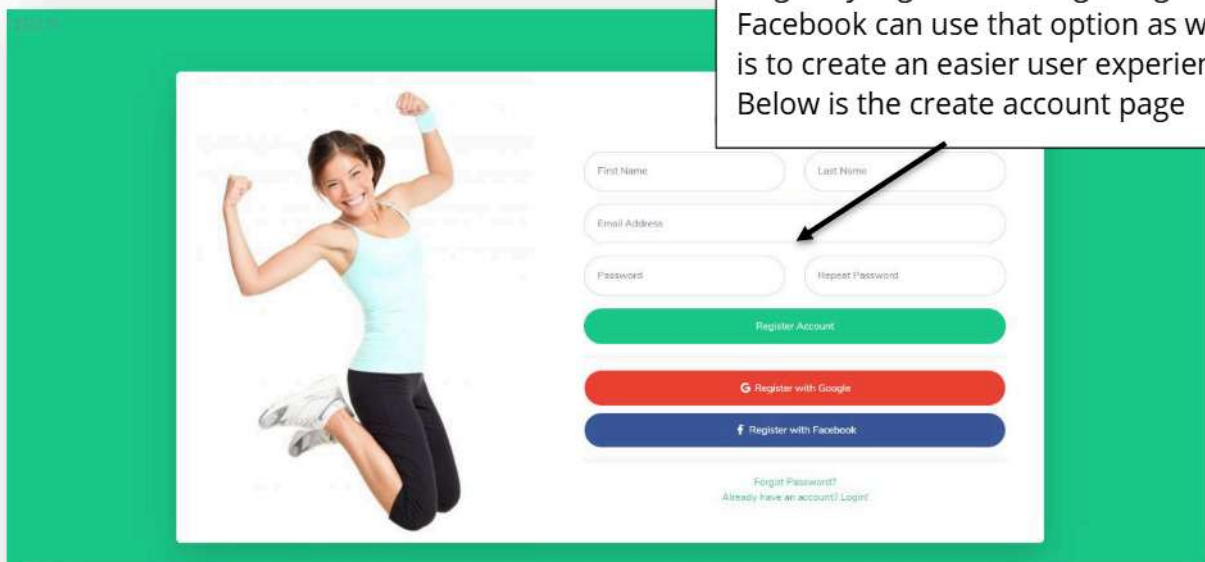
- Fat
- Weight
- Food
- Comparisons and logs for day, week and Month

Log in page



The screenshot shows a 'Create an Account' form on a green background. On the left is a photo of a woman in a light blue tank top and black leggings jumping joyfully. The form on the right has the title 'Create an Account!' and the following fields: 'First Name', 'Last Name', 'Email Address', 'Password', and 'Repeat Password'. Below these fields are three buttons: a green 'Register Account' button, a red 'G Register with Google' button, and a blue 'f Register with Facebook' button. At the bottom of the form, there are links for 'Forgot Password?' and 'Already have an account? Login!'.

This is the login screen; you can use created username and password or if you originally registered using Google or Facebook can use that option as well. This is to create an easier user experience. Below is the create account page



This is a duplicate of the screenshot above, showing the 'Create an Account' form with the woman jumping in the background. It includes the same fields for First Name, Last Name, Email Address, Password, and Repeat Password, along with the Register Account, Register with Google, and Register with Facebook buttons, and the links for Forgot Password? and Already have an account? Login!.

TOKA FITNESS

Dashboard

YOUR HEALTH

Fitness

Diet

SOCIAL

Follow us

Blog

Chat

HELP

Training Videos

Food Plan Videos

How to use Toka Fitness

Get Premium

Buy premium

Billing Address

Full Name
John M. Doe

Email
john@example.com

Address
542 W. 15th Street

City
New York

State
NY

Zip
10001

Payment

Accepted Cards
VISA, Mastercard, American Express

Name on Card
John More Doe

Credit card number
1111-2222-3333-4444

Exp Month
September

Exp Year
2018

CVV
352

Cart

1

Premium £10

Total £10

☒ Shipping address same as billing

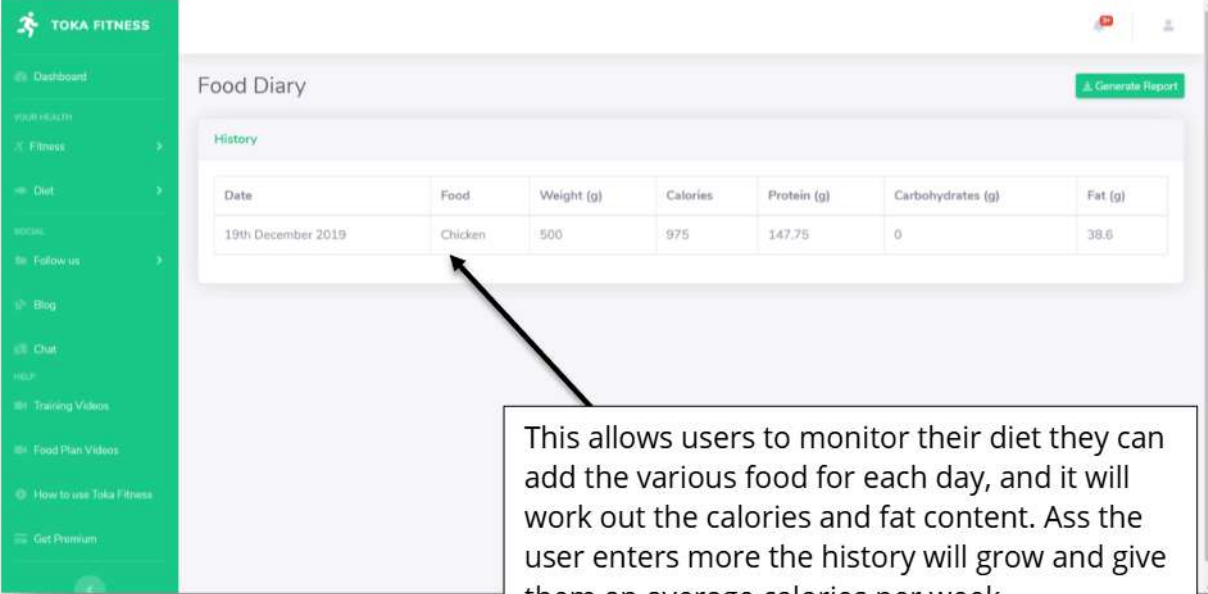
This is the payment page which allows clients to pay for the members sections which gives them access to various extra parts which are not available to standard users

Client information Table



This allows users to view fitness stats to help measure success and to see how their training is progressing.

Food diary is to capture how they monitor the food intakes. Evidence suggests that by viewing what they eat daily will help with keeping their diet in place



TOKA FITNESS

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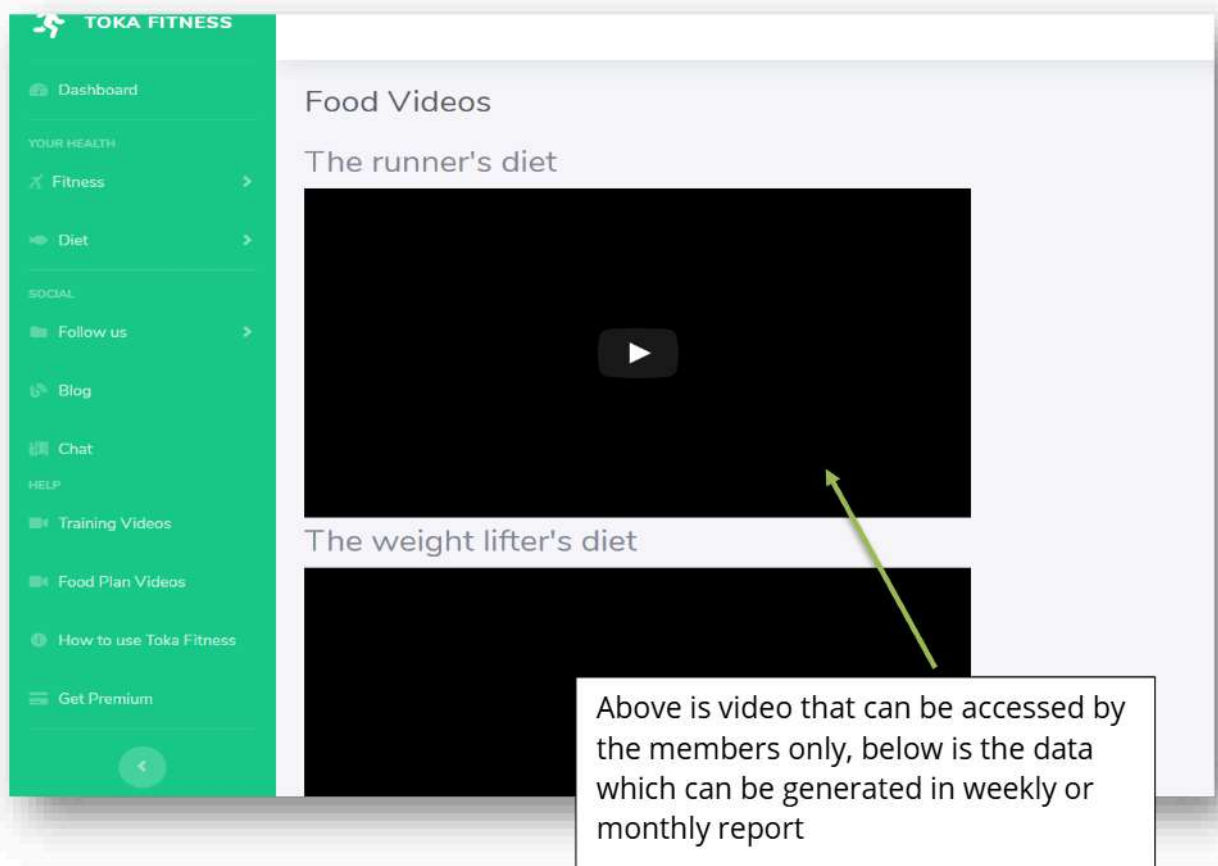
Food Diary

[Generate Report](#)

History

Date	Food	Weight (g)	Calories	Protein (g)	Carbohydrates (g)	Fat (g)
19th December 2019	Chicken	500	975	147.75	0	39.6

This allows users to monitor their diet they can add the various food for each day, and it will work out the calories and fat content. Ass the user enters more the history will grow and give them an average calories per week.



TOKA FITNESS

- Dashboard
- YOUR HEALTH
 - Fitness
 - Diet
- SOCIAL
 - Follow us
 - Blog
 - Chat
- HELP
 - Training Videos
 - Food Plan Videos
 - How to use Toka Fitness
 - Get Premium

Fitness Diary

History

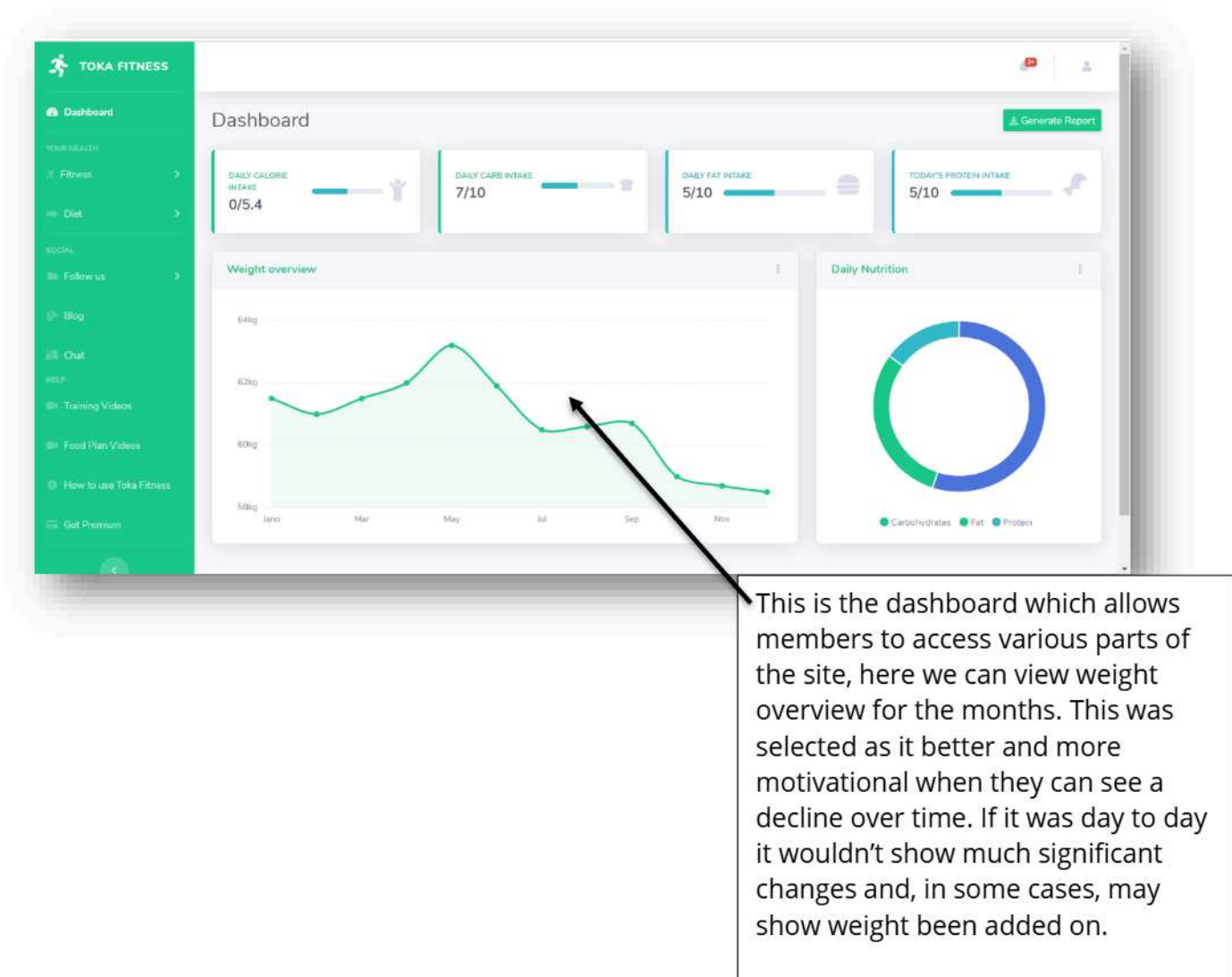
Date	Exercise	Calories burnt	
20th December 2019	Run	300	
20th December 2019	Swim	420	1h
19th December 2019	Walk	142	30m
18th December 2019	Run	300	45m
17th December 2019	Swim	420	1h
17th December 2019	Walk	142	30m
16th December 2019	Run	300	45m
	Swim	420	1h
	Walk	142	30m

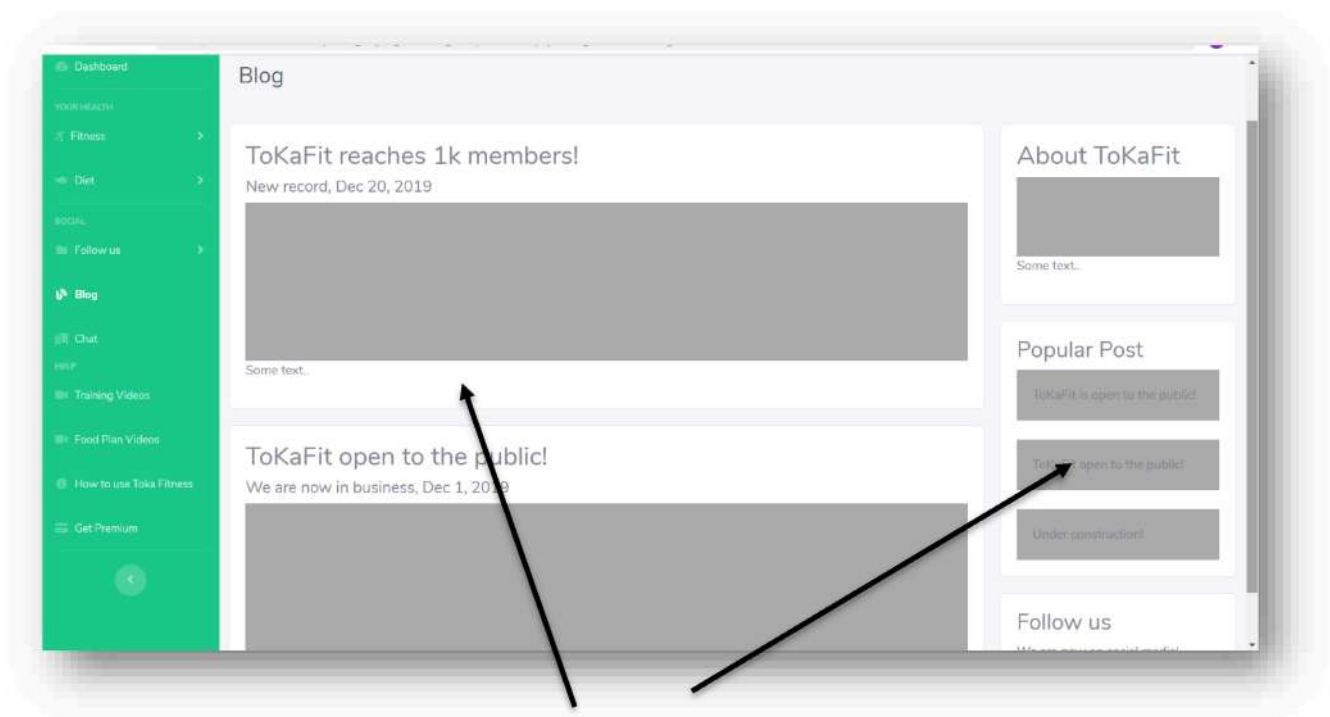
ALERTS CENTER

- December 13, 2019
A new monthly report is ready to download!
- December 7, 2019
You have achieved your monthly goal!

Generate Report

Here the members can run a report for monthly stats which can be downloaded. Also have option to check they have reached their goal for the month





This is the blog page this allows has member and no member's access. The member can see both and return messages to other members.

Also includes popular post and links to Facebook, and twitter accounts.

The screenshot shows the 'Diet Tracker' page of the Toka Fitness application. On the left is a green sidebar with navigation links: Dashboard, YOUR HEALTH (Fitness, Diet, YOUR DIET PAGES: Food Diary, Statistics, Tracker), SOCIAL (Follow us, Blog, Chat), and HELP (Training Videos). The main content area is titled 'Diet Tracker' and contains a form titled 'Add exercise data'. This form has two input fields: 'Food:' and 'Amount (grams):', followed by a 'Submit' button. On the right side of the page, there is an 'ALERTS CENTER' widget showing two notifications: 'December 12, 2018: A new monthly report is ready to download!' and 'December 7, 2018: You have achieved your monthly goal!'. A 'Show All Alerts' link is at the bottom of the alerts center.

This page is used for Diet tracking, the user inputs food eaten and the amount in calories. Once submitted it will keep a record of the different amount of food eaten each day. A monthly report is generated which then can be compared to various element such a protein, carb intake and calories.

The screenshot shows the 'Fitness Tracker' page of the Toka Fitness application. The green sidebar is identical to the Diet Tracker page. The main content area is titled 'Fitness Tracker' and features a green button at the top right labeled 'Import from fitness tracker'. Below this, there are two forms. The first form, titled 'Add exercise data', has a radio button selection for 'Exercise' with options: Running, Walking, Swimming, and Other. It also has a 'Duration (minutes):' input field and a 'Submit' button. The second form, titled 'Add current weight', has a 'Weight (kg):' input field and a 'Submit' button.

The fitness tracker page relies on adding exercise data, this could be for the four most common options. The button at the top allows this you to import the data from a fitness tracker.

Website code

PHP code for member page

```
C:\Users\Administrator\Desktop\ToKaFit\YoKaFit\login.php - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
index.php login.php
1 <?php
2
3 include('config.php');
4 session_destroy();
5 session_start();
6 if (isset($_POST['login'])) {
7     $email = $_POST['email'];
8     $password = $_POST['password'];
9     $query = $connection->prepare("SELECT * FROM users WHERE email
10 $query->bindParam("email", $email, PDO::PARAM_STR);
11 $query->execute();
12
13 $result = $query->fetch(PDO::FETCH_ASSOC);
14
15 if (!$result) {
16     echo '<script>alert("Invalid username and password combina
17 } else {
18     if (password_verify($password, $result['password'])) {
19         $_SESSION['user_id'] = $result['id'];
20         $_SESSION['first_name'] = $result['firstname'];
21         $_SESSION['last_name'] = $result['lastname'];
22         header("Location:index.php");
23         die();
24     } else {
25         echo '<script>alert("Invalid username and password com
26
27
28
29
30 >>
```

This is the PHP code for the login. Link to include another page. Uses a nested if statement

Session set up to do the following:

- makes user data available across the whole website
- makes a temporary file in a server temporary directory which saves session data
- atomically destroys when user close the browser

Super global variable with session data after session started

Select the user details

Binds the parameter to PHP variable in SQL statement

Produces a message if password invalid

```

1 <?php
2
3 include('config.php');
4 session_start();
5
6 if (isset($_POST['register'])) {
7
8     $firstname = $_POST['firstname'];
9     $lastname = $_POST['lastname'];
10    $email = $_POST['email'];
11    $password = $_POST['password'];
12    $repeatpassword = $_POST['repeatpassword'];
13    if ($password != $repeatpassword) {
14        echo '<h1><font color="white"><center>Password and repeat password do not match!</center></font></h1>';
15    } else {
16        $password_hash = password_hash($password, PASSWORD_BCRYPT);
17
18        $query = $connection->prepare("SELECT * FROM users WHERE EMAIL=:email");
19        $query->bindParam("email", $email, PDO::PARAM_STR);
20        $query->execute();
21
22        if ($query->rowCount() > 0) {
23            echo '<h1><font color="white"><center>The email address is already registered!</center></font></h1>';
24        }
25
26        if ($query->rowCount() == 0) {
27            $query = $connection->prepare("INSERT INTO users(FIRSTNAME, LASTNAME,PASSWORD,EMAIL) VALUES (:firstname, :lastname,:password_hash,:email)");
28            $query->bindParam("firstname", $firstname, PDO::PARAM_STR);
29            $query->bindParam("lastname", $lastname, PDO::PARAM_STR);
30            $query->bindParam("password_hash", $password_hash, PDO::PARAM_STR);
31            $query->bindParam("email", $email, PDO::PARAM_STR);
32            $result = $query->execute();
33
34            if ($result) {
35                echo '<h1><font color="white"><center>Your registration was successful!</center></font></h1>';
36            } else {
37                echo '<h1><font color="white"><center>Something went wrong!</center></font></h1>';
38            }
39        }
40    }
41 }
42
43 ?>
44 <!DOCTYPE html>
45 <html lang="en">
46

```

To collect the data from the form, used the PHP super global variable.

Used bindParam with the SQL statement

```

1 <?php
2 session_start();
3
4 if(!isset($_SESSION['user_id'])){
5     header('Location: login.php');
6     exit;
7 }
8
9
10 if(!isset($_SESSION['premium'])){
11     echo 'You need to be a premium member to access this part of the website!';
12 } else {
13
14 }
15 ?>

```

Isset function is used to check whether there is a value entered. If there is no value, it echoes a message back to user to go to the login in page. Then it checks if the value entered is equal to the value in the database in this case

If userID valid it will allow the use to access the training video. In other words, they are a paying customer. Echo a message if they are not

Used bootstrap JS with some of the functionality on the pages. Menu, graphs and login screen.

```
C:\Users\Administrator\Desktop\TokaFit\TokaFit\Fitness_statistics.php - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

368 <!-- End of Page Wrapper -->
369
370 <!-- Scroll to Top Button-->
371 <a class="scroll-to-top rounded" href="#page-top">
372 <i class="fas fa-angle-up"></i>
373 </a>
374
375 <!-- Logout Modal-->
376 <div class="modal fade" id="logoutModal" tabindex="-1" role="dialog" aria-labelledby="exampleModalLabel" aria-hidden="true">
377 <div class="modal-dialog" role="document">
378 <div class="modal-content">
379 <div class="modal-header">
380 <h5 class="modal-title" id="exampleModalLabel">Ready to Leave?</h5>
381 <button class="close" type="button" data-dismiss="modal" aria-label="Close">
382 <span aria-hidden="true"></span>
383 </button>
384 </div>
385 <div class="modal-body">Select 'Logout' below if you are ready to end your current session.</div>
386 <div class="modal-footer">
387 <button class="btn btn-secondary" type="button" data-dismiss="modal">Cancel</button>
388 <a class="btn btn-primary" href="login.php">Logout </a>
389 </div>
390 </div>
391 </div>
392
393 <!-- Bootstrap core JavaScript-->
394 <script src="vendor/jquery/jquery.min.js"></script>
395 <script src="vendor/bootstrap/js/bootstrap.bundle.min.js"></script>
396
397 <!-- Core plugin JavaScript-->
398 <script src="vendor/jquery-easing/jquery.easing.min.js"></script>
399
400 <!-- Custom scripts for all pages-->
401 <script src="js/sh-admin-2.min.js"></script>
402
403 <!-- Page level plugins -->
404 <script src="vendor/chart.js/Chart.min.js"></script>
405
406 <!-- Page level custom scripts -->
407 <script src="js/demo/chart-area-demo.js"></script>
408 <script src="js/demo/chart-pie-demo.js"></script>
409
410 </body>
411 </html>
412
413
414
PHP Hypertext Preprocessor file length: 11,679 lines: 314 Ln: 1 Col: 1 Sel: 0|0 Unix (LF) UTF-8 INS
```

These styles apply to all elements (except h1 and scrollbar style). The scrollbar styling only works on select browsers like chrome.

The Meta viewport tag lets me control the content on different sized screens.

Linked jQuery to the website, so it can be used in the JavaScript code (e.g. to style elements).


```

1 function () {
2   Number.prototype.toHex = function () {
3     var ret = ((this<0?0x8:0)+((this >> 28) & 0x7)).toString(16) + (this & 0xffffffff).toString(16);
4     while (ret.length < 8) ret = '0'+ret;
5     return ret;
6   };
7   Object.prototype.hashCode = function hashCode(o, l) {
8     l = l || 2;
9     var i, c, r = [];
10    for (i=0; i<l; i++)
11      r.push(i*268803292);
12    function stringify(o) {
13      var i,r;
14      if (o === null) return 'n';
15      if (o === true) return 't';
16      if (o === false) return 'f';
17      if (o instanceof Date) return 'd:'+(o+o);
18      i=typeof o;
19      if (i === 'string') return 's:' + o.replace(/([\\\/:])/g, '\\$1');
20      if (i === 'number') return 'n:' + o;
21      if (o instanceof Function) return 'm:' + o.toString().replace(/([\\\/:])/g, '\\$1');
22      if (o instanceof Array) {
23        r=[];
24        for (i=0; i<o.length; i++)
25          r.push(stringify(o[i]));
26        return 'a:' + r.join(':');
27      }
28      r=[];
29      for (i in o) {
30        r.push(i+''+stringify(o[i]))
31      }
32      return 'o:' + r.join(':');
33    }
34    o = stringify(o);
35    for (i=0; i<o.length; i++) {
36      for (c=0; c<r.length; c++) {
37        r[c] = (r[c] << 13) - (r[c] >> 19);
38        r[c] += o.charCodeAt(i) << (r[c] % 24);

```

Hashing Function, using JS allows the password to scramble to increase security.

C:\Users\19741\Documents\Year 13 CS\Unit 3 Projects\Jack_Male\Jack_Male\Project DB - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

main.py inventory_2022.py del_statistics.php package-lock.json training_videos.php blog.php php.xml workspace.xml rosides.xml fitness_tracker.php ESP_Task_V4_21-07-2023.py ProjectDB

```

1 CREATE TABLE [import].[ExerciseLog]
2 (
3   [ExerciseLogID] INT NOT NULL CONSTRAINT PK_ExerciseLogID PRIMARY KEY,
4   [ExerciseID] SMALLINT NOT NULL,
5   [ActivityDate] DATETIME2 NOT NULL,
6   [ExerciseCount] TINYINT NOT NULL,
7   [GymUserID] INT NOT NULL,
8   [FirstInserted] DATETIME2 NOT NULL DEFAULT getdate(),
9   CONSTRAINT [FK_ExerciseLog_Exercise] FOREIGN KEY ([ExerciseID]) REFERENCES [import].[Exercise]([ExerciseID]),
10  CONSTRAINT [FK_ExerciseLog_GymUser] FOREIGN KEY ([GymUserID]) REFERENCES [import].[GymUser]([GymUserID])
11 )
12

```

```

1 <?php
2 session_start();
3
4 require_once "setupDb.php";
5
6 $password = $passwordError = $oldPassword = $oldPasswordError = $confirmPass = $confirmPassError =
7
8 function filterInput($data) {
9     $data = trim($data);
10    $data = stripslashes($data);
11    $data = htmlspecialchars($data);
12    return $data;
13 }
14
15 if ($_SERVER["REQUEST_METHOD"] == "POST") {
16     $oldPassword = filterInput($_POST["oldPassword"]);
17     $password = filterInput($_POST["newPassword"]);
18     $confirmPass = filterInput($_POST["passConfirm"]);
19
20     if (empty($oldPassword)) {
21         $oldPasswordError = "Please enter your old password";
22         $_SESSION["passError"] = $oldPasswordError;
23     }
24     elseif(empty($password)) {
25         $passwordError = "Please enter your new password";
26         $_SESSION["passError"] = $passwordError;
27     }
28     elseif (empty($confirmPass)) {
29         $confirmPassError = "Please confirm your new password";
30         $_SESSION["passError"] = $confirmPassError;
31     }
32     else {
33         if ($password == $confirmPass) {
34             //SQL update statement to change user information
35             $sql = "SELECT passwordHash FROM userinfo WHERE email = ?";
36
37             if ($stmt = $connection->prepare($sql)) {
38                 //binds variables to parameters
39                 $stmt->bind_param("s", $param_email);
40
41                 //set parameters
42                 $param_email = $_SESSION["email"];
43
44                 if ($stmt->execute()) {
45                     $stmt->store_result();
46
47                     $stmt->bind_result($passwordHash);
48                     if ($stmt->fetch()) {
49                         if (password_verify($oldPassword, $passwordHash)) {
50                             //SQL statement below updates data on the database
51                             $sql = "UPDATE userinfo SET passwordHash = ? WHERE email = ?";

```

This function allows the user to reset password, checks the person existing email and binds the two together.

These 3 if statements check if the user entered a value in the input boxes.

This SQL statement selects the current password hash of the user from the database. This is done to compare to the user's old password input (if the input = password hash, then the program continues).

```
52
53         if ($stmt = $connection->prepare($sql)) {
54             //binds variables to parameters
55             $stmt->bind_param("ss", $param_passwordHash, $param_email);
56
57             //parameters are set
58             //converts new password into hash
59             $param_passwordHash = password_hash($password, PASSWORD_DEFAULT);
60             $param_email = $_SESSION["email"];
61
62             if ($stmt->execute()) {
63                 //changes password characters into asterisks
64                 $password = str_repeat(" ", strlen($password));
65                 $_SESSION["password"] = $password;
66                 $_SESSION["passError"] = "";
67             }
68         }
69         $stmt->close();
70     }
71     else {
72         $_SESSION["passError"] = "Old password is not correct";
73     }
74     $connection->close();
75 }
76
77 else {
78     //If the SQL statement didn't work, the resulting phrase will be outputted (in the profile page using session variable)
79     $_SESSION["passError"] = "Sorry, something went wrong, please try again";
80 }
81
82 $stmt->close();
83 }
84 else {
85     $confirmPassError = "Passwords do not match";
86     $_SESSION["passError"] = $confirmPassError;
87 }
88
89 $connection->close();
90 }
91
92 //returns to profile page
93 header("location: profile.php");
94 exit;
95 ?>
```

Testing

Test data	Expected outcome	Reasons for test	Actual outcome
email: a@a password: a	Log in normally	Check if valid inputs work	As expected, the user is logged in
Test data	No log in	Check if erroneous inputs not in database won't work	As expected, user is not logged in
Empty inputs			
Email: ab@abc Password: 1111	No log in	Check if valid inputs won't work if not existing in database	As expected, user is not logged in
Empty inputs	No log in	Check if invalid inputs won't work	As expected, user is not logged in due to lack of inputs

Sign up

Password: 4444 confirm password: 4444			
Title: Mr First name: qq Last name: sss Email: oooo@a Phone no: abc Password: ss s Confirm password: sss	Account is not made	To check if website recognises invalid phone no. (not only being numbers)	As expected, no account is made

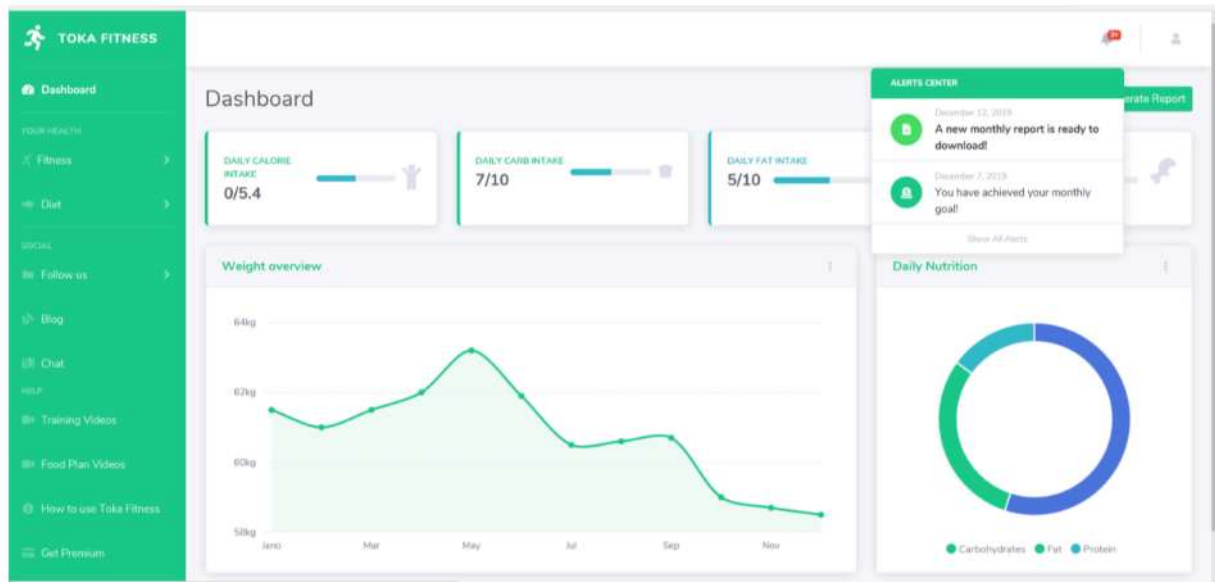
Title: Mr First name: qqg Last name: sss Email: oooo@a Phone no: abc Password: sss Confirm password: sss2	Account is not made	To check if website checks if password = confirmed password	As expected, no account is made
Title: Mr First name: qqg Last name: sss Email: oooo Password: sss Confirm password: sss2	Account is not made	To check if website checks validity of email (syntax)	As expected, account is not made

Element to be tested	Expected result	Test Data	Expected result from test data	Actual result
Header tab navigation system	When user clicks on buttons system directs user to right webpage.	n/a	n/a	System successfully directs user to requested webpage.
Sign up validation	When user inputs data, the system checks the data and outputs an error if the input isn't inappropriate for the field: First name and Last name must only contain letters and not numbers or special characters the email field must contain the "@" and "." Characters There must be at least one character	First name: Chris 555 Last Name: Jones Email: ChrisJones@gmail.com (email already registered) Email (2nd test):ChrisJones2@gmail.com	Error popup and form cannot be submitted	System directs user to an error page if the email is not unique However doesn't validate other values such as first name, last name, email syntax, postcode and password and submits the data into the database. System outputs an error message when no data is inputted into the form.
Dashboard	View dashboard	Imported data from user input	Viewed dashboard and all graphs	No issues all graphs could be seen
Graphs	View graphs, bar, pie charts	Imported data	Viewed graphs with relevant statistics	No issues
Video	View video	YouTube Videos	Select and view video for fitness and diet	No issues

Fitness Diary	View tabular data	Date: December Exercise: Run, Swim and Walk Calories: range Distance: Range	View data in tabular format with range of dates, exercises, calories and distance	No issues
Diet Tracker	Inputs various food and weight in grams	Chicken 400 grams	Submit data for tracking	No issues

Test Results

Outcome for dashboard



Test outcome Fitness Diary

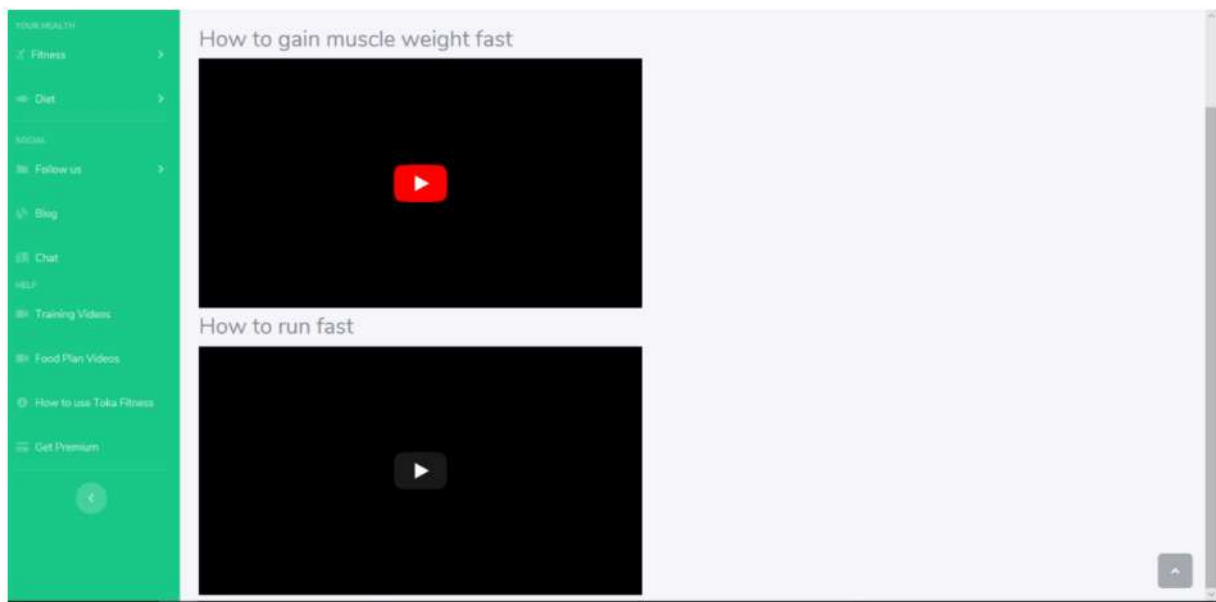
The screenshot displays the TOKA FITNESS Fitness Diary. The left sidebar is identical to the dashboard. The main content area is titled 'Fitness Diary' and shows a 'History' table. An 'ALERTS CENTER' pop-up is also present on the right. The table contains the following data:

Date	Exercise	Calories burnt	
20th December 2019	Run	300	
20th December 2019	Swim	420	1h
19th December 2019	Walk	142	30m
18th December 2019	Run	300	45m
17th December 2019	Swim	420	1h
17th December 2019	Walk	142	30m
16th December 2019	Run	300	45m
15th December 2019	Swim	420	1h
14th December 2019	Walk	142	30m
	Run	300	45m

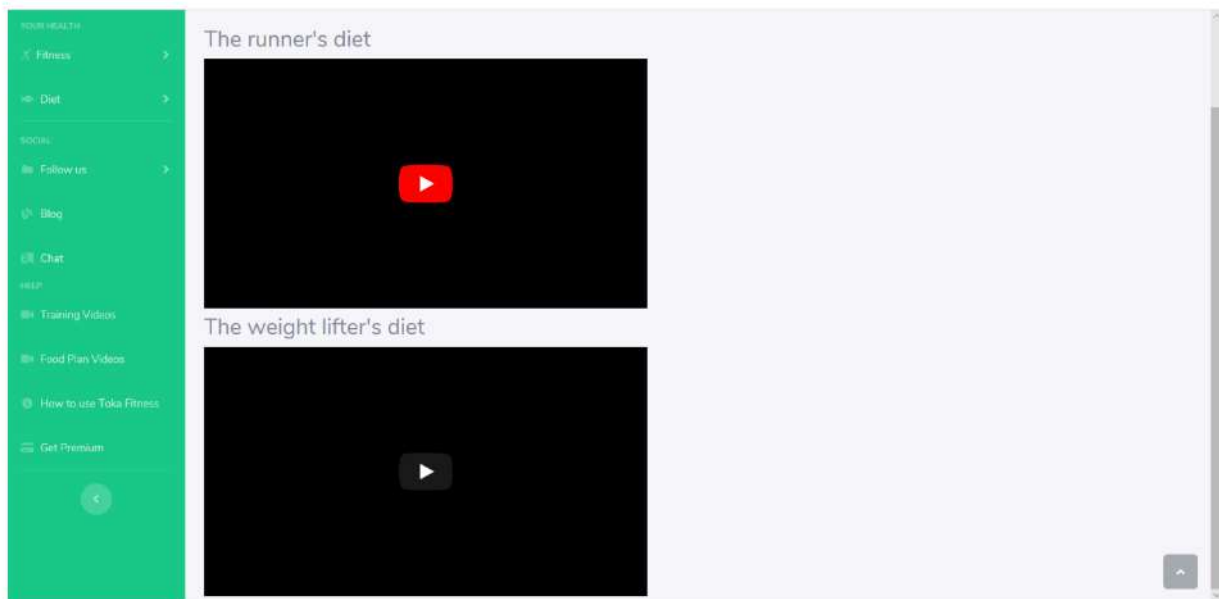
Test outcome Diet Tracker

The screenshot shows the Toka Fitness web application. On the left is a green sidebar with a logo and a list of navigation items: Dashboard, YOUR HEALTH (Fitness, Diet), SOCIAL (Follow us, Blog, Chat), and HELP (Training Videos, Food Plan Videos, How to use Toka Fitness, Get Premium). The main content area is titled 'Diet Tracker' and contains a form titled 'Add exercise data'. The form has two input fields: 'Food:' with the value 'Chicken' and 'Amount (grams):' with the value '400'. A 'Submit' button is at the bottom of the form.

Test outcome Training Video



Test outcome Diet Videos



Profile info editing:

Test data	Expected outcome	Reasons for test	Actual outcome
No inputs	Website asks for inputs	Check if lack of inputs is detected	As expected, error message is displayed
Valid inputs (e.g. abc for first name)	Website accepts input and info is changed	Check if valid inputs are accepted	As expected, error message is displayed
Old password is incorrect	Website tells user, the input is wrong	Check if incorrect data is detected	As expected, the error message is displayed
Old pass is correct, New password = ccc Confirm password = asdf	Website tells user passwords don't match	Check if code works as it should	As expected, the user is notified of the exact problem

Unit to test Sign-In Assumption

Unit to test	Sign-In
Assumption	The webpage displayed the sign-in input screen and wait for user's action.
Input	user ID, user password, user information

Expect output

Register successfully with proper input, register unsuccessfully with improper, and make alter the same time.

Pass	Function fit the requirement.
------	-------------------------------

Fail and probable error

- (1) Register still success with improper input. Error related input checking.
- (2) Register success with proper input but no related data show in data base. Error related to connection with data base.

```

1  <?php
2
3  include('config.php');
4  session_start();
5
6  if (isset($_POST['register'])) {
7
8      $firstname = $_POST['firstname'];
9      $lastname = $_POST['lastname'];
10     $email = $_POST['email'];
11     $password = $_POST['password'];
12     $repeatpassword = $_POST['repeatpassword'];
13     if($password != $repeatpassword) {
14         echo '<h1><font color="white"><center>Password and repeat password do not match!</center></font>
15     } else {
16         $password_hash = password_hash($password, PASSWORD_BCRYPT);
17
18         $query = $connection->prepare("SELECT * FROM users WHERE EMAIL=:email");
19         $query->bindParam("email", $email, PDO::PARAM_STR);
20         $query->execute();
21
22         if ($query->rowCount() > 0) {
23             echo '<h1><font color="white"><center>The email address is already registered!</center></font>
24         }
25
26         if ($query->rowCount() == 0) {
27             $query = $connection->prepare("INSERT INTO users(FIRSTNAME, LASTNAME, PASSWORD_HASH, EMAIL) VALUES (:firstname, :lastname, :password_hash, :email)");
28             $query->bindParam("firstname", $firstname, PDO::PARAM_STR);
29             $query->bindParam("lastname", $lastname, PDO::PARAM_STR);
30             $query->bindParam("password_hash", $password_hash, PDO::PARAM_STR);
31             $query->bindParam("email", $email, PDO::PARAM_STR);
32             $result = $query->execute();
33
34             if ($result) {
35                 echo '<h1><font color="white"><center>Your registration was successful!</center></font>
36             } else {
37                 echo '<h1><font color="white"><center>Something went wrong!</font></center></h1>';
38             }
39         }
40     }
41 }

```

There were a few issues when trying to run the code, added a \$ to the if which made it variable, removed this and program worked

```

21
22     if ($query->rowCount() > 0) {
23         echo '<h1><font color="white"><center>The email address is already registered!</center></font>
24     }

```



```

1 <?php
2
3 include('config.php');
4 session_start();
5
6 if (isset($_POST['register'])) {
7     $firstname = $_POST['firstname'];
8     $lastname = $_POST['lastname'];
9     $email = $_POST['email'];
10    $password = $_POST['password'];
11    $repeatpassword = $_POST['repeatpassword'];
12    if ($password != $repeatpassword) {
13        echo '<h1><font color="white"><center>Password and repeat password
14    </h1>';
15    } else {
16        $password_hash = password_hash($password, PASSWORD_BCRYPT);
17
18        $query = $connection->prepare("SELECT * FROM users WHERE EMAIL=:email");
19        $query->bindParam("email", $email, PDO::PARAM_STR);
20        $query->execute();
21
22        if ($query->rowCount() > 0) {
23            echo '<h1><font color="white"><center>The email address is already
24        </h1>';
25        }
26
27        if ($query->rowCount() == 0) {
28            $query = $connection->prepare("INSERT INTO users(FIRSTNAME, LASTNAME,
29            (:firstname, :lastname, :password_hash, :email) VALUES
30            $query->bindParam("firstname", $firstname, PDO::PARAM_STR);
31            $query->bindParam("lastname", $lastname, PDO::PARAM_STR);
32            $query->bindParam("password_hash", $password_hash, PDO::PARAM_STR);
33            $query->bindParam("email", $email, PDO::PARAM_STR);
34            $result = $query->execute();
35
36            if ($result) {
37                echo '<h1><font color="white"><center>Your registration was successful!</center></font>
38            </h1>';
39            } else {
40                echo '<h1><font color="white"><center>Something went wrong!</font></center></h1>';
41            }
42        }
43    }
44 }
45

```

Another problem was the semi colon at the end of the statement. This was corrected and loaded the login page

Another problem which was hard to detect was missed curly brackets. Added the missing brackets and the code interpreted fine

```

24 </h1>';
25 }
26
27 if ($query->rowCount() == 0) {
28     $query = $connection->prepare("INSERT INTO users(FIRSTNAME, LASTNAME,PASSWORD,EMAIL) VALUES
29     (:firstname, :lastname, :password_hash, :email)");
30     $query->bindParam("firstname", $firstname, PDO::PARAM_STR);
31     $query->bindParam("lastname", $lastname, PDO::PARAM_STR);
32     $query->bindParam("password_hash", $password_hash, PDO::PARAM_STR);
33     $query->bindParam("email", $email, PDO::PARAM_STR);
34     $result = $query->execute();
35
36     if ($result) {
37         echo '<h1><font color="white"><center>Your registration was successful!</center></font>
38     </h1>';
39     } else {
40         echo '<h1><font color="white"><center>Something went wrong!</font></center></h1>';
41     }
42 }
43 }
44 }
45

```